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Evaluation of Different Forklift Battery Systems Using Statistical Analysis and Discrete Event Simulation

Abdulhameed Alshaebi[1], Husam Dauod[1], Joseph Weiss[2], and Sang Won Yoon[1]

[1]Department of Systems Science and Industrial Engineering
The State University of New York at Binghamton
Binghamton, NY 13902

[2]The Raymond Corporation
Greene, NY 13778

Abstract

This paper presents a comparative performance analysis of two different forklift battery systems in the cold warehouse environment. Forklift battery is a key element that significantly affects forklift efficiency; i.e., a high-performance battery minimizes forklift downtime and contributes to better energy consumption. Due to the wide range of available battery technologies, battery choice becomes a challenging decision. In this research, two different batteries are compared: 1) Lead Acid Batteries (LAB) and 2) Lithium Ion Batteries (LIB). Actual warehouse data are obtained from two identical reach forklift trucks equipped with LAB and LIB, and various statistical analyses are conducted to study battery performance in terms of energy efficiency during different truck events such as lifting and traveling. Then, a discrete event simulation model is developed to investigate each battery performance under different scenarios. The results indicate that the lithium ion battery minimizes energy consumption and process time by providing trucks with opportunity charging, i.e. charging during breaks, and enhancing their traveling acceleration and lifting speed.

Keywords
Data Analytics, Discrete Event Simulation, Forklift battery

1. Introduction
Forklifts are critical resources that directly influence the overall efficiency of any warehouse or manufacturing facility. Therefore, the design and operational features of forklifts have been studied by many researchers to maximize their performance. Forklift battery is a key element that significantly affects forklift efficiency. A high-performance battery minimizes forklift downtime and contributes to better energy efficiency. In addition, other features of the battery such as size, weight, and temperature endurance are important since they affect the performance of the forklift. However, due to the wide range of available battery technologies, battery choice becomes a challenging decision, hence, statistical analysis and mathematical modeling techniques are needed to choose the best battery option.

The lead acid battery system is the most common technology used in material handling applications. However, previous research indicates that the lithium ion battery system, which is a relatively new technology, might be a better system in terms of energy efficiency [1]. The lithium ion battery system has also several competitive features such as light weight, small size, ability to operate over a wide range of temperature, and being more environmental friendly [3]. To the best of our knowledge, these battery systems have never been compared in an actual warehouse environment, hence, this paper evaluates their performance in terms of energy efficiency, truck productivity, and truck downtime. In addition, this paper investigates the performance of both battery systems in a cold storage zone since it is expected that low operating temperature influences the battery performance [2].

In this study, a 700 Ah lithium ion battery and 1240 lead acid battery are installed in reach forklift trucks. Batteries are equipped with sensors to collect data related to voltage, current, and travel and lift speed. The reach trucks are

monitored, and the data are collected over a period of five weeks. Then, a statistical analysis is conducted to analyze and compare the performance of each battery, and a discrete simulation model is developed to demonstrate the performance of both systems under different warehouse scenarios. In the simulation model, battery parameters are incorporated and 20 trucks scenario is analyzed to calculate 3 performance measures:1) truck productivity which is the average number of pallet movements per operating hour; 2) energy efficiency which is the ratio of useful energy (i.e., energy delivered from battery to truck) to the rated capacity; and 3) truck downtime.

The remainder of this paper is organized as follows: Section 2 presents relevant literature; Section 3 explains the research methodology; Section 4 illustrates and discusses results; and Section 5 provides the conclusions and future work of this research.

2. Literature Review

Energy storage systems have been extensively studied in the literature. While most studies focused on battery characteristics such as energy density, cycle life, and operating temperature, few studies have addressed battery performance in actual operating environments. In this paper, the performance of lead acid and lithium ion batteries is evaluated in warehouse settings by analyzing battery data collected from electric reach trucks. Lead acid battery is the most battery used for electric vehicles [4, 5]. However, since the lead acid battery has a low energy density, different battery technologies have been introduced such as nickel cadmium and lithium ion [5]. A comparison between these technologies was conducted in terms of energy and power density and the findings indicated that lithium ion is the most efficient battery. The reason is that it provides higher energy per time unit, which in turn allows vehicles to accelerate faster [4]. Lithium ion has also been used in aircraft applications for its advantages that include light weight and high energy density. However, the main disadvantage of this technology is related to its high initial cost [6].

Batteries are used over a wide range of applications, but their performance varies according to the application and the operating environment. Temperature is the most influencing factor on batter's performance; while high temperature reduces battery cycle life; low temperature minimizes its capacity [7]. A previous study performed on three batteries: lead acid, lithium ion and nickel metal hydride investigated the temperature impact on cycle life. Results indicated that the lithium ion battery provides better cycle life at extreme temperatures, but its less efficient than other batteries at medium temperatures [7]. In addition, it was shown that lithium ion is more efficient with respect to self-discharge, that is the amount of energy loss in idle status [8]. A comparative study in the literature has also showed that that lithium ion is more sensitive than lead acid to temperature, meaning that lithium ion does not operate well over a wide range of temperature [8]. In general, battery performance does not only depend on the battery design and specification, but also on the operating environment [9], thus, it is possible to find some variation in the battery performance among different applications. In this paper, lead acid and lithium ion battery systems are compared in a cold warehouse setting. Statistical analysis and discrete event simulation are conducted to choose the optimal battery system for reach forklift trucks based on multiple performance measures.

3. Methodology

In this research, lead acid and lithium ion batteries are installed on reach trucks to compare their performance in a cold warehouse setting. Data have been collected over a period of 5 weeks and a comparative analysis has been done to determine the best technology for reach truck. Three main comparative measures are used: 1) energy efficiency, 2) truck productivity, and 3) truck downtime.

3.1 Data Collection

Two reach trucks are selected for this study; one is equipped with a new 700 Ah lithium ion battery (i.e., LI truck) and the other is equipped with 1240 Ah lead acid battery (i.e., LA truck). These trucks are instrumented with data collection devices to monitor the battery state and truck work for a five-week period with a sample rate 10 times per second. A sample of the collected data is shown in Table 1.

Table 1: Sample of collected data

Timestamp	Voltage - V	Current - A	Reach	Retract	Travel Speed MPH	Lift Speed FPS
7/10 1:03:01	39.37	3.35	0	0	1.35	0.14
7/10 1:03:01	39.39	7.83	0	0	1.33	0.13

The research assumptions are summarized as follows:

- Because lithium ion battery can be charged at a high C rate at any available time (i.e., there is no need to change the battery) and lead acid battery must be changed whenever it reaches low state of charge (SOC), the discharged cycle for batteries is defined as: 1) lithium ion discharge cycle is the period of consuming 80 % of the nominal battery capacity; and 2) lead acid discharge cycle is the time period from when a battery is fully charged until it gets changed.
- Battery internal resistance may change over time, therefore, to simplify energy waste calculations, the internal resistance for both batteries is assumed to be constant.
- Since many operators drive the same truck in different shifts, the effect of drivers on energy efficiency is not considered.

3.2 Statistical Analysis

Data preprocessing and statistical analysis have been used in this research to convert the collected data into useful information. The objective is to determine the following KPIs for each truck: 1) total pallets movements, 2) total discharge cycles, 3) energy delivered per discharge cycle, 4) total operating hours, and 5) total downtime. Truck productivity is calculated by dividing the total pallet movements over the total operating hours. Energy efficiency is the ratio of delivered energy to rated capacity per discharge cycle. Figure 1 illustrates how pallet movements are counted; each pick-drop event (i.e., consecutive red and blue points) represents one pallet movement.

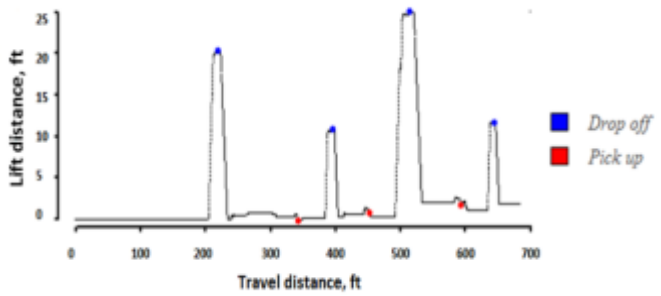


Figure 1: Illustration of the pallet movement analysis

3.3 Discrete Event Simulation

Discrete event simulation provides a flexible approach to model complicated systems and the interactions between individuals and their environment. It has been utilized in a wide range of applications such as manufacturing, healthcare, and warehousing operations. The main advantage of simulation is the ability to evaluate different scenarios without the need to change the real system. In this study, the performance of lithium ion and lead acid batteries are investigated under what-if scenarios using Demo3D. The developed model simulates the operations of 20 reach trucks in a busy warehouse environment. To model the actual work of trucks in the warehouse, the collected travel and lift distances are organized and used as an input for the model. Therefore, the model will import the travel and lift distance for each pallet movement.

Battery parameters such as voltage and current are incorporated into the simulation model, and the state-of-charge is estimated according to the voltage curve when the truck is idle. The lead acid battery is replaced when SOC reaches 20%. The battery replacement time for the LA truck follows a uniform distribution of (8,20) minutes. For the lithium ion truck, the battery is charged automatically when parked. Lithium ion and lead acid scenarios are run for six consecutive shifts, which is equivalent to two days, for 10 replications. After that, performance measures are calculated from the simulation results. Figure 2 is a screenshot from the simulation model.



Figure 2: Simulation model of warehouse operations

4. Results and Discussion

In this section, the findings of each comparative measure are presented separately and the simulation results are discussed. These comparative measures are the most important measures for the warehouse management and they include: 1) truck productivity, 2) energy efficiency, and 3) truck downtime.

4.1 Truck Productivity

In the warehouse, reach trucks are used to move pallets within different areas. Each pallet movement has a specific travel distance, where the truck travel from point to another, and lift distance, where the truck lifts the pallet from one shelf level to the other. The number of pallets moved at the end of the shift is influenced by the assigned workload, i.e. total distance travelled. Therefore, more pallet movements don't necessarily imply better productivity unless the workload difference between the trucks is not significant. Table 2 shows the number of pallets moved, the total operating hours, and the average number of pallets moved per hour for each truck.

Table 2: Productivity of both trucks

Truck Name	Total Pallet Movements	Total Operating Time (hours)	Average Pallets Moved per hour
Lead acid Truck	3,991	221	18.1
Lithium ion Truck	4,008	190	21.1

The results in Table 2 indicate that the lithium ion truck performed on average 3 pallets more than lead acid truck per hour. The Histograms of the travel and lift distances assigned for each truck are shown in Figures 3 and 4.

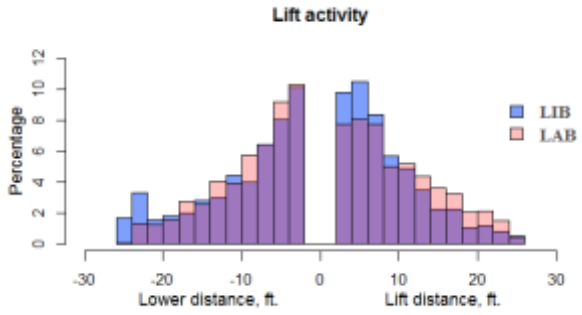


Figure 3: Lift distance distribution

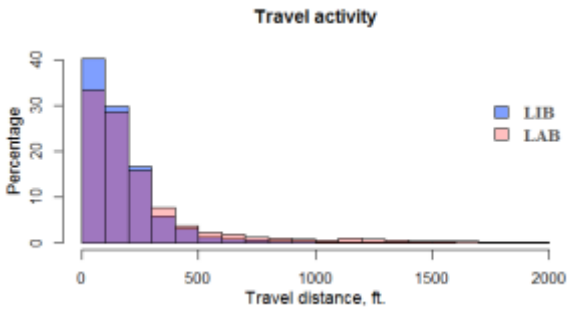


Figure 4: Travel distance distribution

A statistical test is conducted for travel and lift activities with 95% confidence interval. The results indicate that the workload difference is significant since the p-value is 0.001(less than 0.05). Therefore, the workload difference has to be examined before determining which truck is more productive.

4.2 Energy efficiency

Energy efficiency is the energy delivered from the battery to the truck as a percentage of its rated capacity. The energy delivered is calculated for each discharge cycle in Ah. In total, 32 discharge cycles have occurred for both batteries over the study period. Figure 5 shows the amount of energy delivered by the battery in each discharge cycle, while Figure 6 shows the energy efficiency (i.e., practical capacity) per each discharge cycle.

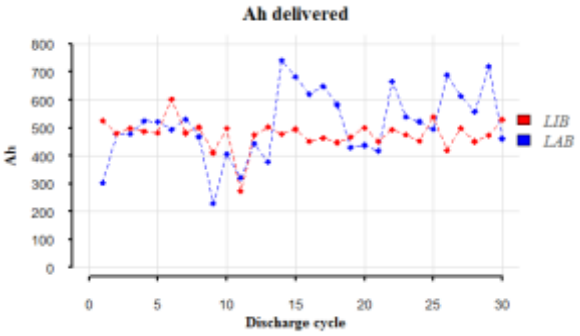


Figure 5: Ah delivered per discharge cycle

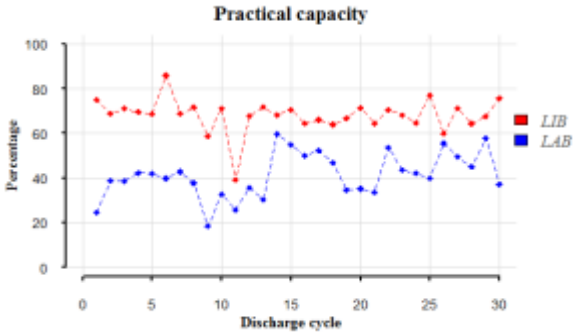


Figure 6: Practical capacity per discharge cycle

The results shown in Figure 5 indicate that the lithium ion and the lead acid batteries can both deliver on average around 500 Ah. However, since their rated capacities are different, the percentage of usable energy should be calculated. Figure 6 illustrates the percentage of usage for each battery relative to its rated capacity. While lithium ion uses about 70% of its capacity, lead acid uses only around 45% of its capacity. Therefore, the lithium ion is 55% better than lead acid. In addition, the heat generated by the lithium ion battery is only half of the heat generated by the lead acid battery. The reason is that the internal resisitance of lithuim ion is less than the internal resistance of lead acid.

4.3 Truck downtime

Lithium ion battery can be charged while being in the truck; thus, no battery replacement processes are needed. However, the lead acid battery needs to be replaced whenever its SOC reaches 20%. The time required to replace the lead acid battery and travel to and from the battery room (i.e., Non-productive time) is illustrated in Figure 7. On average, 1.5 hours per truck per week could be saved if the lithium ion battery is used instead of the lead acid battery.

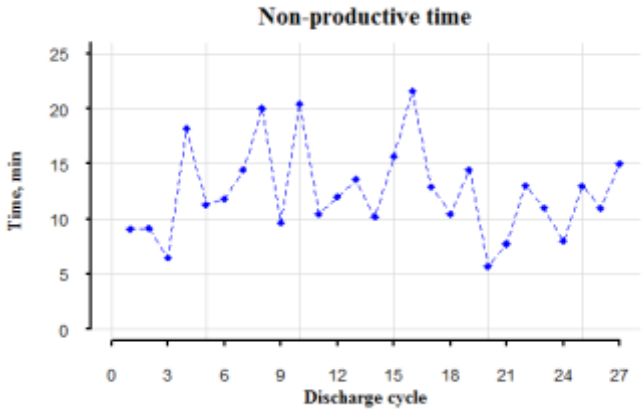


Figure 7: Non-productivity per discharge cycle

4.4 Simulation model

The two scenarios, 20 lithium ion trucks and 20 lead acid trucks, are tested for six consecutive shifts with 10 replications. The results show that around 25 labor hours are spent in lead acid battery management each week, while lithium ion battery does not require this time. In addition, the lithium ion battery improves productivity and energy consumption by 50% and 13% respectively. Table 3 summarizes the simulation results

Table 3: Summary of simulation results

Battery	Downtime per Week per Truck (hours)	Energy Efficiency %	Truck Productivity per Hour
LAB	26	46	18.5
LIB	0	73	21

5. Conclusion and Future Work

Forklift battery technology selection is a challenging task for forklift designers and warehouse managers as it directly affects the forklift efficiency and subsequently the overall performance. In this paper, the promising lithium ion battery system is tested and compared with the currently used lead acid battery. The comparison is conducted using statistical analysis and simulation techniques, and it focuses mainly on truck productivity, truck downtime, and energy efficiency. The findings indicate that lithium ion adoption saves around 25 labor hours per week for a fleet of 20 reach trucks since it does not need to be changed for charging purposes. Lithium ion battery also delivers more than 70% of its rated capacity, while only 46% of the rated capacity can be extracted from lead acid battery. Finally, lithium ion truck shows higher productivity rate with more than 10% increase in number of pallets moved per operating hour. However, in order to between these two batteries, the initial cost should be considered since the lithium ion is more expensive than the lead acid. Therefore, financial feasibility analysis is suggested as future work.

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